

B.Tech IV Year I Semester (R20) Regular Examinations December/January 2024

AUTOMOBILE ENGINEERING

(Mechanical Engineering)

Time: 3 hours

Max. Marks: 70

PART – A
(Compulsory Question)

- 1 Answer the following: (10 X 02 = 20 Marks)
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| (a) Define Automobile. Give the typical specifications of an automobile. | 2M |
| (b) What are the functions of piston and crank shaft? | 2M |
| (c) State the advantages and limitations of electronic ignition system. | 2M |
| (d) What is the function of carburettor? | 2M |
| (e) What is power steering? | 2M |
| (f) What is the objective of vehicle suspension? | 2M |
| (g) Give detailed account of a tyre specification. | 2M |
| (h) Write the advantages of ABS braking system over conventional systems. | 2M |
| (i) List the name of automobile battery components. | 2M |
| (j) What is fuel cell? | 2M |

PART – B

(Answer all the questions: 05 X 10 = 50 Marks)

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| 2 | (a) Discuss the methods of vehicle construction in detail. | 6M |
| | (b) Write about oil pumps and filters. | 4M |
| OR | | |
| 3 | (a) Explain with neat sketch the principle of crankcase ventilation. | 4M |
| | (b) What is the purpose of lubrication in an automobile engine? Explain any one with neat sketch. | 6M |
| 4 | (a) With a neat sketch explain the magneto coil type ignition system. | 5M |
| | (b) Describe the working of Electronic fuel injection system with a neat sketch. | 5M |
| OR | | |
| 5 | (a) Explain working principle of Mechanical fuel pump with neat sketch. | 5M |
| | (b) Explain MPFI system for petrol engine with neat sketch. | 5M |
| 6 | (a) How is wheel alignment done in automobile? Explain. | 4M |
| | (b) Explain with neat sketch the independent rear suspension system. | 6M |
| OR | | |
| 7 | (a) Explain any one type of steering gear box with neat sketch. | 5M |
| | (b) What is a shock absorber? Explain any one in detail. | 5M |
| 8 | (a) Draw the neat sketch of construction of tyre and name the parts. | 5M |
| | (b) Sketch the arrangement of pneumatic braking system used in automobiles and explain. | 5M |
| OR | | |
| 9 | (a) Write the wheel types and specifications. Sketch and explain the construction of a disc wheel. | 5M |
| | (b) Compare the working of vacuum brakes and hydraulic brakes. | 5M |
| 10 | (a) Describe with neat sketch the Active Suspension System. | 5M |
| | (b) Explain the construction and lay out of Hybrid Vehicles | 5M |
| OR | | |
| 11 | (a) With the help of a neat diagram describe Electronic Brake Distribution. | 5M |
| | (b) Write short note on Global Positioning System (GPS). | 5M |
